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A Comparative Analysis of Budget Forecasting Methods: A Systematic Literature Review Covering the 1983–2024 Period

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ABSTRACT

This study systematically analyzes 69 peer-reviewed works comparing budget forecasting methods. It explores methodological evolution, geographic distribution, and evaluation trends. Four phases of development are identified: statistical methods, diversification, machine learning, and deep learning. A division emerges between traditional and next-generation techniques. Geographically, 43% of studies focus on the United States, while developing economies remain underrepresented. In evaluation, MAPE, RMSE, and MAE dominate, with directional errors largely neglected. Findings show that optimal method choice depends on context, supporting a pluralistic, context-sensitive approach rather than universal reliance on a single forecasting method.

1 | Introduction

Public budgets, which lie at the core of fiscal management systems, function both as instruments of economic policy and as frameworks for managing public resources. The forecasts used in budget preparation determine the government's future fiscal capacity, spending priorities, and economic orientation. The forecasting process is shaped by a wide range of factors, including macroeconomic indicators, political preferences, institutional capacity, and technical methodologies. In this respect, budget forecasts go beyond being a merely technical element of fiscal planning, constituting a strategic component of the policy-making process.

This strategic position necessitates that budget forecasts be prepared in a realistic, reliable, and accurate manner. Efficient allocation of scarce public resources and the successful coordination of public–private sector planning are only possible through a budget process grounded in robust forecasts (Krause and Corder 2007). Similarly, the timely and effective implementation of fiscal policy measures (Botrić and Vizek 2012) and the design and execution of sound fiscal policies (Derrick 2002; Danninger 2005) depend critically on the accuracy of forecasts. Moreover, forecasts serve as key indicators for economic agents: various financial indicators, particularly interest rates, are influenced by these forecasts; households and financial market participants base their long-term decisions on them; and central banks that pursue inflation targeting rely on such data to guide their policy decisions.

Accurate forecasts serve as a fundamental tool for determining the extent to which public services and programs can be financed (Auld 1970; Bretschneider et al. 1989). The success of forecasting is particularly critical for the efficient financing of large future public expenditures (Auld 1970), as multiyear expenditure forecasts represent an estimate of the cost of providing a certain level of public service (Schroeder 2007). Furthermore, establishing a robust link between long-term planning and implementation depends on the accuracy of

Summary

- This study provides a systematic analysis of 69 peer-reviewed studies comparing budget forecasting methods, identifying four distinct phases of methodological evolution from basic statistical models to the deep learning era.
- The findings reveal that no single forecasting method is universally superior; instead, the optimal choice critically depends on contextual factors such as the economic environment, data quality, and the forecast horizon.
- A significant finding is the pronounced geographical concentration of research, with 43% of studies focused on the United States, leaving forecasting methodologies in developing economies substantially underrepresented.
- The field is further characterized by a hegemony of certain performance metrics (MAPE, RMSE, and MAE), which, while allowing for comparison, largely neglect the critical policy dimension of directional forecast errors.
- Consequently, the study advocates for a shift away from the quest for a single best model toward a more pluralistic, context-sensitive, and potentially hybrid approach to budget forecasting.

medium- and long-term forecasts (Asher 1978; Jena 2006; Zakaria and Ali 2010; Fedotov 2017). Having sufficient information about available resources enables more effective management of long-term programs and projects within the budget process, while careful evaluation of this information guides project and program selection, helping to determine what is feasible in the medium term (Jena 2006). In this context, forecasts related to a country's key sectors provide a scientific foundation for strategic planning (Fedotov 2017).

The realistic preparation of budget forecasts is also a critical prerequisite for the government to achieve its planned objectives (Khan and Anwar Hussain 2018; Bretschneider and Gorr 1992; Agostini 1991). As observed in modern economies, a government's ability to respond effectively to any crisis is directly influenced by the quality of the financial information available to it, and consequently, by the accuracy of forecasts (Willoughby and Guo 2008). In other words, the success of forecasting shapes public perception regarding the effectiveness of a government's fiscal policy (Leal et al. 2008). Therefore, major errors in budget forecasting negatively affect the outcomes of governmental programs and policies, potentially leading to policy failure (Srinivasan and Misra 2020). Accordingly, it can be argued that forecast accuracy plays a significant role in the budgetary policy process by contributing to transparency, accountability, and fiscal discipline (McNab et al. 2005).

As a natural consequence of this strategic importance, efforts to enhance the accuracy of budget forecasts have remained a continuous focus in the field of public finance. Two principal approaches emerge in this regard. The first involves identifying the causes of past forecasting errors and developing remedial mechanisms. This approach aims to eliminate recurring error patterns by examining in detail the structural, institutional, economic, and political factors influencing the forecasting process. The second approach focuses on improving the methods and models used in the preparation of forecasts. Such methodological advancement encompasses both the refinement of existing techniques and the implementation of new modeling approaches, with the goal of strengthening the analytical capacity of the forecasting process.

The literature contains numerous studies that focus on these two approaches. These studies have compared the performance of various forecasting techniques across different country contexts and datasets. The models employed range widely, from simple time-series methods based on trend analysis to complex econometric models, and from deterministic approaches to artificial intelligence (AI) and machine learning (ML)-based algorithms. While simpler methods offer advantages such as ease of implementation and low data requirements, more complex techniques, despite requiring greater computational power and extensive data sources, possess the potential to capture dynamic relationships. This diversity indicates that there is no single "ideal" method for budget forecasting; rather, the choice of method depends on factors such as economic conditions, data quality, institutional capacity, and policy priorities.

The existing literature does not provide a homogeneous assessment of the success of methods used in budget forecasting, due to variations in datasets, time periods, countries, and model types. Forecasts produced across a broad spectrum, from simple time-series techniques to complex ML approaches, exhibit varying performance in different contexts. This situation necessitates a systematic evaluation of which methods perform better under specific conditions. Such an assessment would not only contribute to the advancement of academic knowledge but also serve as a practical guide for institutions, policymakers, and analysts involved in the budget process regarding method selection and model development. In response to this need, the present study offers a comprehensive review of the existing literature, providing a comparative evaluation of different models and techniques.

The critical need for advancing budget forecasting accuracy is further underscored by recent large-scale efforts to define the research priorities for the entire field of public budgeting and finance. In a comprehensive agenda-setting study, McDonald et al. (2024) engaged a wide community of academics and practitioners to identify the most pressing research needs. Notably, their findings, derived from a ranked-choice voting process, highlighted "Data and Methods" as a top-tier priority, explicitly calling for research that improves "the accuracy of budget and revenue forecasting." This consensus positions methodological investigations, such as the present study, at the forefront of the field's agenda. Our systematic review directly addresses this call by synthesizing the comparative performance of a wide array of forecasting techniques over a four-decade period. By doing so, this study aims not only to contribute to academic knowledge but also to provide a practical, evidence-based guide for practitioners in selecting and developing forecasting models, thereby bridging the gap between academic inquiry and the operational needs identified by the community of practice.

The paper is structured as follows. First, the dataset used in the analysis is introduced, detailing its scope, coverage, and key characteristics. Second, the methodology is explained. Third, the state of the literature on budget forecasting methods is examined, with a focus on comparative findings and methodological diversity. Finally, the conclusion section summarizes the main findings and outlines implications for future research and policy design.

2 | Data Set

The dataset constructed in this study encompasses the academic literature in which budget forecasting methods have been comparatively evaluated. In creating the dataset, a series of selection criteria was established to clearly define the scope and ensure comparability. First, only articles published in peer-reviewed journals, books, book chapters, and scientific conference proceedings were included. This choice aims to ensure that the methods employed and the results obtained have been scrutinized according to academic standards and are highly reliable. Reports, technical notes, or institutional assessments that have not undergone a peer-review process were excluded from the dataset.

Second, since the focus of this study is on method comparison, budget forecasting studies that employ only a single method were not included. The dataset consists of research in which multiple forecasting methods were applied within the same study and their performance compared. This approach allows for the direct observation and analysis of performance differences across methods.

Third, it was not sufficient for the included studies to provide only forward-looking forecasts; these forecasts were required to be compared with actual outcomes for the corresponding periods. Studies that did not perform such comparisons were excluded from the dataset, as they could not objectively demonstrate the success of the forecasting methods. This requirement is critical for ensuring a robust assessment of methodological performance, which constitutes the primary motivation of the study.

In some studies, forecasts cover both past periods, for which actual outcomes are available, and future periods, which have not yet occurred. In such cases, only the periods with available actual data were included in the dataset. For example, in a study with a 10-year forecasting horizon where actual outcomes are available for only the first five years, the dataset represents only the five-year segment. This approach ensures that method performance is measured solely over verifiable periods.

Finally, terminological consistency was maintained in the naming of methods within the study. In different studies, minor variations in model structure or configuration often led to the same method being referred to by different names; in the dataset, these were consolidated under a single heading. For instance, various variants of artificial neural networks (ANNs), even when labeled with different prefixes in the original studies, were grouped under the ANN category in the dataset. This standardization enables accurate and meaningful comparisons across methods.

The primary motivation for constructing this dataset is to enable an objective evaluation of the comparative performance of budget forecasting methods and to conduct a systematic analysis of findings in the literature. In this way, the study aims to provide clearer and more reliable insights into the conditions under which different methods yield superior results, both for the academic community and for practical applications.

3 | Methods

The purpose of this study is to examine the distribution, temporal trends, and interrelationships of methods and metrics employed in the budget forecasting literature. While classical bibliometric analyses are typically conducted through citation networks, co-authorship collaborations, or journal co-occurrence mapping (Passas 2024), this study adopts a different perspective. Instead of focusing on citation relationships or collaborative networks, the analysis emphasizes the frequency of method applications, the prevalence of metric preferences, the characteristics of training datasets, the length of forecasting horizons, and the interconnections among methods. Thus, although the research shares certain features with bibliometrics, it is primarily designed within the framework of statistical analysis and network analysis. The analytical process consists of three stages: (i) descriptive statistical analysis, (ii) trend analysis, and (iii) network analysis.

3.1 | Descriptive Statistical Analysis

Descriptive statistical analysis aims to summarize the fundamental characteristics of collected data and provide an overall framework for the research field (Cooksey 2020). Such analyses are frequently used in bibliometric studies to reveal the quantitative structure of the literature (Castelino et al. 2024). In this study, descriptive analyses include the frequency of method applications, the annual distribution of publications, the geographical (country-level) distribution of research, the extent of metric preferences, and the length of forecasting horizons. The findings are visualized through graphs, offering a general overview of methodological diversity in the literature.

3.2 | Trend Analysis

Trend analysis is a method used to examine temporal changes in a given phenomenon and to identify underlying patterns. It is frequently employed to illustrate the evolution of research domains and methodological transformations, thereby contributing to the understanding of the developmental trajectory of a field (Li 2025). In this study, trend analysis was conducted to investigate the intensity of method applications over time. The results were visualized with line graphs, highlighting methods that gained prominence during different periods.

3.3 | Network Analysis

Network analysis is a powerful method for examining structures composed of nodes (e.g., methods, authors, or metrics) and edges that represent the connections among them. This approach is widely used to investigate the structural characteristics and methodological diversity of scientific domains (Campbell et al. 2024), offering valuable insights into the interactions among different components of the literature.

In this study, network analysis was performed using Gephi software, resulting in three distinct network structures: method–method, method–year, and method–metric relationships. Nodes represented methods, years, and metrics, while edges denoted their co-occurrences. This analysis enabled the visualization of which methods were prominent in specific periods, which metrics they were evaluated with, and which methods tended to be employed jointly.

4 | Results

The table below provides a comprehensive summary of empirical studies on budget forecasting methods. It systematically classifies results based on different country contexts, data frequencies, time periods, and types of methods, thereby highlighting the relative performance of various approaches. In this way, the fragmented findings in the literature regarding the success of budget forecasting methods can be evaluated from a holistic perspective, allowing for a comparative assessment of which methods perform better or worse under specific conditions. Within this framework, the table not only serves as an inventory of previous research but also makes methodological diversity and performance differences, central to the motivation of this paper, explicit, laying the groundwork for subsequent analyses.

Academic interest in the comparative performance of budget forecasting methods has shown a steady and remarkable increase over the past four decades. The reviewed literature reveals not only a numerical growth in studies but also a significant evolution in terms of scope, methodology, and geographical diversity. Early contributions (e.g., Downs and Rocke 1983; Litterman and Thomas 1983) were largely confined to advanced economies (particularly the United States) and relied primarily on relatively simple time-series models (ARMA and ARIMA). Over time, however, research has expanded to include developing countries as well as far more complex econometric and AI-based models. This expansion is a clear indication that the pursuit of greater forecasting accuracy has taken on a global and interdisciplinary character. Table 1 provides a systematic summary of this rich and dynamic body of work, detailing the scope, methods, and key findings of the 68 comparative studies that form the basis of the present analysis.

From a methodological perspective, the literature exhibits a clear trend toward increasing complexity and diversity. While the 1980s and early 1990s were dominated by classical econometric approaches such as ARIMA, VAR, and ECM, the 2000s marked the entry of ML and deep learning (DL) algorithms into the field. ANN, SVM, and, more recently, cutting-edge methods such as LSTM networks and XGBoost have been incorporated into the forecasters' toolkit (e.g., Yang et al. 2023; M. Khalifa et al. 2024). At the same time, hybrid models and innovative approaches such as MIDAS, which address the issue of frequency mismatch (Ghysels and Ozkan 2012, 2015), have gained prominence in performance comparisons. This evolution demonstrates that researchers have not only sought to improve traditional methods but have also pushed the boundaries of forecasting accuracy by leveraging data abundance and advances in computational power.

The expansion in geographical coverage is also noteworthy. Whereas early studies were almost entirely United States-centered, recent years have witnessed significant contributions from a wide range of economies, including China, Turkey, Brazil, South Africa, India, and numerous Eastern European countries, as illustrated in Table 1. This diffusion underscores that budget forecasting is a universal issue in public finance and highlights that method performance may vary across different economic structures, institutional arrangements, and levels of data quality.

Therefore, the findings in the literature are not homogeneous; a method that delivers the "best" performance in one country or period may yield only modest results in another context. This heterogeneity further underscores the need for a systematic assessment of which methods perform better under specific conditions and reinforces the primary motivation of this study.

First, the distribution and frequency of forecasting methods employed in the comparative studies under review were analyzed over time. This analysis provides an objective lens through which to observe the evolution of methodological trends, technological capabilities, and academic interest in the budget forecasting literature. Such a historical perspective is crucial for understanding which methods dominated in particular periods, how the popularity of specific models has shifted, and when pioneering contributions introduced innovations to the field.

The analysis of Figure 1 demonstrates that the methodological trajectory of the budget forecasting literature can be characterized as a four-stage evolution. The first stage, emerging in the 1980s, was dominated by basic statistical methods. During this period, traditional time series and econometric models such as ARIMA, VAR, and various regression techniques constituted the main focus of academic studies. In subsequent years, greater diversification and sophistication in modeling techniques became evident, with more complex and theoretically grounded approaches such as BVAR, ECM, and MIDAS being introduced. This phase marked a period in which researchers sought to enhance forecasting power by making model structures more dynamic and by integrating different types of data. By the 2010s, a notable methodological shift had occurred: ML algorithms such as ANN and SVM began to feature regularly in the literature, bringing an interdisciplinary dimension to the field. This transition was driven by the growing availability of large datasets, advances in computational capacity, and increasing interest in modeling nonlinear relationships. Finally, since the 2020s, the literature has entered the era of DL and hybrid models. Cutting-edge algorithms such as LSTM and XGBoost have become frequently tested in comparative analyses.

TABLE 1 | Literature review on the comparative performance of models in budget forecasting.

Author(s)	Country	TO	TYR	Obs. freq.	For. per.	Methods	Best	Worst	Metric(s)
Downs and Rocke (1983)	USA	61	1946–1978	Annual	1946–1979	ARMA; MA; RW	ARMA	RW	None
Litterman and Thomas (1983)	USA	64	1967–1982	Quarterly	1971–1982	VAR; COF; OFC	COF	OFC	MAE; PE
Nazmi and Jane (1985)	USA	44	1970–1980	Quarterly	1981–1983	ARMA; RGR	ARMA	RGR	RMSE
Steinnes and Wong (1985)	USA	156	1970–1984	Monthly	1983–1984	ARIMA; VAR	ARIMA	VAR	PE
Sexton (1987)	USA	462	1974–1980	Annual	1980–1983	ARIMA; Drift; COF	COF	Drift	MSE
Nazmi and Leuthold (1988)	USA	44	1970–1980	Quarterly	1981–1982	RGR; ARIMA	ARIMA	RGR	RMSE; MAE; ME
Fullerton (1989)	USA	16	1978–1981	Quarterly	1982–1985	ARIMA; COF; OFC	COF	ARIMA	RMSE; PE
Frank (1990)	USA	48	1982–1985	Monthly	1986	MA; ES; HW; ARIMA; CurveFit	MA	CurveFit	MAPE
Frank and Gerasimos (1990)	USA	60	1983–1987	Monthly	1988	MA; ES; HW; GAF; COF; OFC	HW	OFC	RMSE; MAPE; APE; PE
Baghestani and McNown (1992)	USA	100	1955–1979	Quarterly	1980–1987	ARIMA; VAR; ECM; OFC	ARIMA	VAR	ME; MAE; RMSE
Duncan et al. (1993)	USA	600	1972–1986	Annual	1979–1986	MSKF; C-MSKF	C-MSKF	MSKF	MAPE; MPE
Gianakis and Howard (1993)	USA	72	1983–1990	Monthly	1990	RGR; MA; HW; ES; ARIMA; GAF;	ARIMA	HW	MAPE
Frank and Wang (1994)	USA	18	1975–1992	Annual	1990–1992	RGR; ES;	ES	RGR	MAPE
Grizzle and Klay (1994)	USA	700	1975–1985	Annual	1980–1985	LOCF; MA; Mean; ES; RGR; CurveFit	CurveFit	MA	MAPE
Otrok and Charles (1997)	USA	58	1983–1997	Quarterly	1990–1997	BVAR; OFC	BVAR	OFC	RMSE; MAE
Cirincione et al. (1999)	USA	99	1990–1996	Qua, Mon	1995–1996	MA; ES; ARIMA; RGR;	ES	MA	MAPE
Favero and Marcellino (2005)	Panel	30	1981–1995	Biannual	1996–2002	ARMA; VAR; SCM; MCM; COF	ARMA	VAR; MCM	RMSE; MAE
Voorhees (2006)	USA	104	1975–2000	Quarterly	1995–2000	ANN; OFC	ANN; OFC	ANN	PE
Kong (2007)	USA	17	1985–2001	Annual	1995–2001	OFC; RGR; LRT; Drift; Mean	RGR	Mean	MAPE
Pérez (2007)	Panel	136	1970–2003	Quarterly	1994–2003	RGR; COF; RW; OFC	COF	RW	RMSE
Rudzkis and Mačiulaitytė (2007)	Lithuania	28	1998–2005	Quarterly	2003–2005	RGR; ECM	RGR	ECM	MAPE
Silvestrini et al. (2008)	France	108	1996–2004	Monthly	2002–2004	ARIMA; OFC	ARIMA	OFC	RMSE; MAE; MAPE
Chatagny and Soguel (2012)	Switzerland	62	1945–2006	Annual	1979–2006	ARIMA; OFC	ARIMA	OFC	PE
Lu et al. (2009)	China	9	1990–1998	Annual	1999–2001	ANN; SVM; GM	SVM	ANN	MAPE

(Continues)

TABLE 1 | (Continued)

Author(s)	Country	TO	TYR	Obs. freq.	For. per.	Methods	Best	Worst	Metric(s)
Pedregal and Pérez (2010)	Panel	288	1984–2007	Ann, Qua, Mon	2001–2007	MIX; RW	MIX	RW	RMSE
Corvalão et al. (2010)	Brazil	85	1995–2001	Monthly	2001	ECM; ARIMA	ECM	ARIMA	MAPE
Hájek and Olej (2010)	Czechia	1808	2003–2006	Annual	2003–2006	ANN; SVM; RGR;	SVM	RGR	RAE; RRSE
Krol (2010)	USA	100	1979–2003	Quarterly	2004–2009	VAR; BVAR; RW	BVAR	RW	RMSE
Onorante et al. (2010)	Panel	372	1963–2016	Monthly	1994–2006	AR; MIX; ECM; RW	MIX	RW	RMSE; DM
Cooke and McIntosh (2011)	USA	28	1970–1997	Annual	1998–2011	ARIMA; HW; COF; Drift; OFC	COF	Drift	RMSE; MAE
Liu and Rufei Zhang (2011)	China	11	1999–2009	Annual	1999–2009	GM; ES; COF	COF	ES	MAPE
Li-xia et al. (2011)	China	10	1990–1999	Annual	1993–2003	ANN; SVM	SVM	ANN	MAPE
Botrić and Vizek (2012)	Croatia	100	1995–2008	Quarterly	2007–2008	Drift; RW; ARIMA; RGR; ECM	RW; ECM	ARIMA; RGR	MAPE
Ghysels and Ozkan (2012)	USA	176	1968–2011	Quarterly	1990–2011	MIDAS; AR; ADL; COF	MIDAS	AR	RMSE
Koirala (2012)	Nepal	192	1997–2012	Monthly	1997–2012	HW; RGR; SARIMA; Drift	SARIMA	HW	MAPE; MPE
Asimakopoulos and Joan Paredes (2013)	Panel	116	1991–2012	Quarterly	2008–2011	MIDAS; Mean; OFC	MIDAS	Mean	RMSE
Zhijun (2013)	China	32	1980–2011	Annual	2008–2011	ANN; ARMA; SVM	ANN	ARMA	MAPE; MSPE
Zhang (2014)	China	33	1980–2012	Annual	1980–2012	SVM; OFC	SVM	OFC	RMSE; RME
Ghysels and Ozkan (2015)	USA	57	1956–2012	Annual	1993–2012	MIDAS; AR; ADL; COF	MIDAS	AR	RMSE
Nandi and Muntasir Chaudhury (2015)	Bangladesh	101	2004–2012	Monthly	2004–2013	SARIMA; HWSMA; HWSAA; OFC	HWSMA	SARIMA	MPE; MSE; MAPE
Alamdari et al. (2016)	Iran	44	1963–2006	Annual	2007–2009	ANN	None	None	RMSE
Chimilila (2017)	Tanzania	182	2000–2015	Monthly	2014–2015	ARMA; COF; GARCH; MA	COF	MA	RMSE; MAE
Shahnazarian and Martin Solberger (2017)	Sweden	110	1993–2014	Quarterly	2010–2014	BVAR; MIDAS; ARIMAX	BVAR	ARIMAX	ME; MAE; RMSE
Ticona et al. (2017)	Brazil	144	2000–2012	Monthly	2013–2014	ANN; GAL; COF; OFC	COF	OFC	MAPE; RE
Makananisa and Erero (2018)	South Africa	36	2009–2017	Quarterly	2015–2017	HW; SARIMA; RGR	HW	SARIMA	ME; RMSE; MAE; MPE; MASE
Sabaj and Kahveci (2018)	Albania	55	2005–2016	Quarterly	2011–2015	RW; HW; ECM; VAR; BVAR; SARIMA; COF; OFC	COF	OFC	RMSE; MAE; MAPE
Streimikiene et al. (2018)	Pakistan	378	1985–2016	Monthly	2017	AR; ARIMA; VAR	ARIMA	AR	RMSE; MAE; MAPE
Buxton et al. (2019)	USA	770	1999–2018	Quarterly	2015–2018	ARIMA; MLP; LSTM	MLP	ARIMA	MAPE

(Continues)

TABLE 1 | (Continued)

Author(s)	Country	TO	TYR	Obs. freq.	For. per.	Methods	Best	Worst	Metric(s)
Erdoğdu and Yorulmaz (2019)	Türkiye	108	2006–2014	Monthly	2015–2018	RW; SARIMA; BATS; OFC	BATS	RW	RMSE; MAE; MAPE; ME; MPE; MASE
McShea and Cordes (2019)	USA	32	1977–2008	Annual	2009–2014	AR; VAR; BVAR; RW	BVAR	RW	RMSE; MAE; MAPE
Molapo et al. (2019)	South Africa	56	1998–2012	Quarterly	2012–2015	BVAR; ARIMA; ES	BVAR; ES	ARIMA	RMSE; MAE
Yılmaz (2019)	Türkiye	108	2006–2014	Monthly	2015–2017	ARIMA; OFC	ARIMA	OFC	MPE
Alhassan et al. (2020)	Nigeria	144	1981–2016	Quarterly	2017–2019	KSVR; W-KSVR; PCA-KSVR; PCA-W-KSVR	PCA-W-KSVR	W-KSVR; PCA-KSVR	MAPE; MAE; MSE; RMSE
Damane (2020)	Lesotho	92	1993–2016	Quarterly	2016–2017	AR-MIDAS; OFC	AR-MIDAS	OFC	RMSE; MAE; MAPE
Ofori and Abel Fumey (2020)	Ghana	180	2002–2016	Monthly	2017–2019	ARIMA; HW	ARIMA	HW	RMSE; MAPE
Ünsal et al. 2020)	Türkiye	11	2006–2016	Annual	2017–2019	GM; OFC	GM	OFC	None
Chung et al. (2022)	USA	116	1994–2014	Annual	1994–2014	MA; ES; ARIMA; ANN; KNN; CART	ARIMA	CART	MAPE; MSE
Ghysels et al. (2022)	USA	41	1958–2012	Annual	1999–2018	MIDAS; AR; ADL; COF; RW	MIDAS	RW	RMSE
Lahiri and Yang (2022)	USA	174	2006–2021	Qua, Mon	2020–2021	MIDAS; LASSO; RW; CGB; OFC	CGB	RW	RMSE; MAE; MPE; MAPE
Mukherjee and Bhattacharya (2023)	Bharat	49	2011–2023	Quarterly	2021–2023	VAR; SARIMA	VAR	SARIMA	MAPE; RMSE
Thayyib et al. (2023)	Bharat	44	2017–2022	Monthly	2021–2022	TBATS; ES; ANN; COF	COF	ANN	RMSE; MAE; MAPE
Yang et al. (2023)	Panel	31	1990–2020	Annual	2015–2020	ARIMA; ETS; SVR; XGBOOST; CNN; LSTM; BiRNN; ASRNN; GRU	GRU	ARIMA	MAE; RMSE; MAPE
M. Khalifa et al. (2024)	Egypt	20	2019–2023	Ann, Qua	2019–2023	ARIMA; LSTM; RF; XGBOOST; KNN; Naïve Bayes	LSTM	Naïve Bayes	MAPE; MSE; RMSE
Larson and Overton (2024)	USA	1005	1991–2015	Ann, Qua, Mon	2016–2017	ARIMA; ES; KNN; ANN; XGBOOST; LRT; LOCF; Mean; Drift	KNN; LRT; Drift	ANN; Mean	MAPE
Xie (2024)	China	30	2002–2021	Annual	2019–2021	MLR; ANN	ANN	MLR	MAE; MAPE
Wong and La (2024)	Australia	71	1992–2009	Quarterly	2009–2019	AR; RW; RF; LASSO; RGR; XGBOOST; ANN	AR; LASSO	RGR	RMSE; MAE

(Continues)

TABLE 1 | (Continued)

Author(s)	Country	TO	TYR	Obs. freq.	For. per.	Methods	Best	Worst	Metric(s)
Chung et al. (2025)	USA	120	2009–2019	Monthly	2017–2019	HW; ARIMA; KNN; GRNN; LOCF	HW	ARIMA	MAPE
Şengüler and Kara (2025)	Türkiye	216	2005–2022	Monthly	2023–2024	HW; LSTM; Prophet; ARIMA; COF	SARIMA	LSTM; Prophet	MAPE
Larson and Overton (2025)	USA	1005	1991–2015	Ann, Qua, Mon	2016–2017	ARIMA; ES; RGR; KNN; ANN; XGBOOST; Drift; Mean; LOCF	Drift; LOCF	ANN; Mean	MAPE

Abbreviations: ADL, Augmented Distributed Lag; ANN, Artificial Neural Network; AR, Autoregressive; ARIMA, Autoregressive Integrated Moving Average; ARIMAX, AutoRegressive Integrated Moving Average with Exogenous Inputs; ARMA, Autoregressive Moving-Average; ASRNN, Attention-Based Sequential Recurrent Neural Network; BATS, Exponential Smoothing State Space Model with Box–Cox Transformation; BiRNN, Bidirectional Recurrent Neural Network; BVAR, Bayesian Vector Autoregression; CART, Classification and Regression Tree; CGB, Component-Wise Gradient Boosting; C-MSKF, Cross-Sectionally Adjusted Multi-State Kalman Filter; CNN, Convolutional Neural Network; COF, Combining of Forecasts; ECM, Error Correction; ES, Exponential Smoothing; GAF, General Adaptive Filtering; GAL, Genetic Algorithms; GARCH, Generalized AutoRegressive Conditional Heteroskedasticity; GM, Gray Model; GRNN, Generalized Regression Neural Network; GRU, Gated Recurrent Unit; For. per., Forecasted Period; HW, Holt-Winters; HWSMA, Holt-Winter Seasonal Multiplicative Approach; HWSAA, Holt-Winter Seasonal Additive Approach; LASSO, Least Absolute Shrinkage and Selection Operator; LOCF, Last Observation Carried Forward; LRT, Linear Trend; LSTM, Long Short Term Memory Network; MA, Moving Average; MCM, Multi-Country Structural Model; MIDAS, Mixed-Data Sampling; MIX, Mixed-Frequency; MLP, Multi-Layer Perceptron; MLR, Multiple Linear Regression; MSKF, Multi-State Kalman Filter; OFC, Official Forecasts; Obs. freq., Observation Frequency; PCA, Principal Component Analysis; RF, Random Forest; RGR, Regression; RW, Random Walk; SARIMA, Seasonal Autoregressive Integrated Moving Average; SCM, Single-Country Structural Model; SVM, Support Vector Machine; SVR, Support Vector Regression; TBATS, Trigonometric Seasonality—Box–Cox Transformation—ARMA Errors—Trend and Seasonal Components; TO, Total observations used to train the model; TYR, Training data's years range; VAR, Vector Autoregression; W-KSVR, Wavelet with Kernel Support Vector Regression; XGBOOST, Extreme Gradient Boosting.

In particular, the emergence of complex hybrid models supported by PCA represents a new stage in which researchers strategically combine feature engineering with advanced algorithms to maximize model performance.

The overall assessment of this methodological evolution reveals that the field of budget forecasting is not static but rather a dynamic area of research, constantly interacting with developments in neighboring disciplines such as econometrics, operations research, and AI. The journey from simple time-series models to complex neural networks reflects both the methodological maturity and the accumulated intellectual capital of the field. At the same time, however, the dense methodological diversity illustrated in Figure 1 suggests the presence of a "methodological inflation" in the literature, a phenomenon that increasingly complicates the task of selecting an optimal method within a given context for policymakers and practitioners.

Second, the frequency with which different forecasting methods are used together within the same study has been examined. This analysis makes it possible to understand which methods researchers most frequently evaluate side by side when conducting performance comparisons, and thus which methodological combinations have gained wider acceptance.

The connections in Figure 2 indicate which pairs of methods are most frequently tested together in comparative studies in the literature. The analyses clearly show that traditional methods are often used both among themselves and in conjunction with next-generation techniques for comparison purposes. For instance, ARIMA is frequently compared not only with its own derivatives, such as SARIMA, which incorporate seasonal variations, but also with ML-based methods, such as ANN or SVM. Similarly, official agency forecasts and various regression techniques appear to serve as a benchmark for almost all methodological classes. In particular, when evaluating the performance of ML algorithms (XGBoost, LSTM, and SVM), comparisons with these traditional methods emerge as a widespread trend. It is also noteworthy that variations within the same model family (e.g., VAR vs. BVAR or AR vs. ARIMA) are frequently analyzed together. This suggests that researchers aim to understand how different specifications or enhancements within a given model family, such as Bayesian approaches or seasonal adjustments, affect forecasting performance. In sum, this visual network structure reveals that comparative analyses in the budget forecasting literature follow a certain systematic pattern: the validity and contribution of a new method are most often assessed against established and widely accepted traditional methods, while the relative strengths and weaknesses of different methodological families are empirically evaluated through these typical comparisons.

Third, the changing trends in the popularity of specific forecasting methods over time have been examined. This analysis makes it possible to objectively observe the shifts in academic interest toward particular methodologies and the dominance cycles of certain methods within the literature. The graph displays the changing frequencies of use over the years for the six most frequently compared methods in the analyzed studies.

The findings derived from Graph 1 clearly reveal shifts in academic trends. Official forecasts were frequently used as benchmark

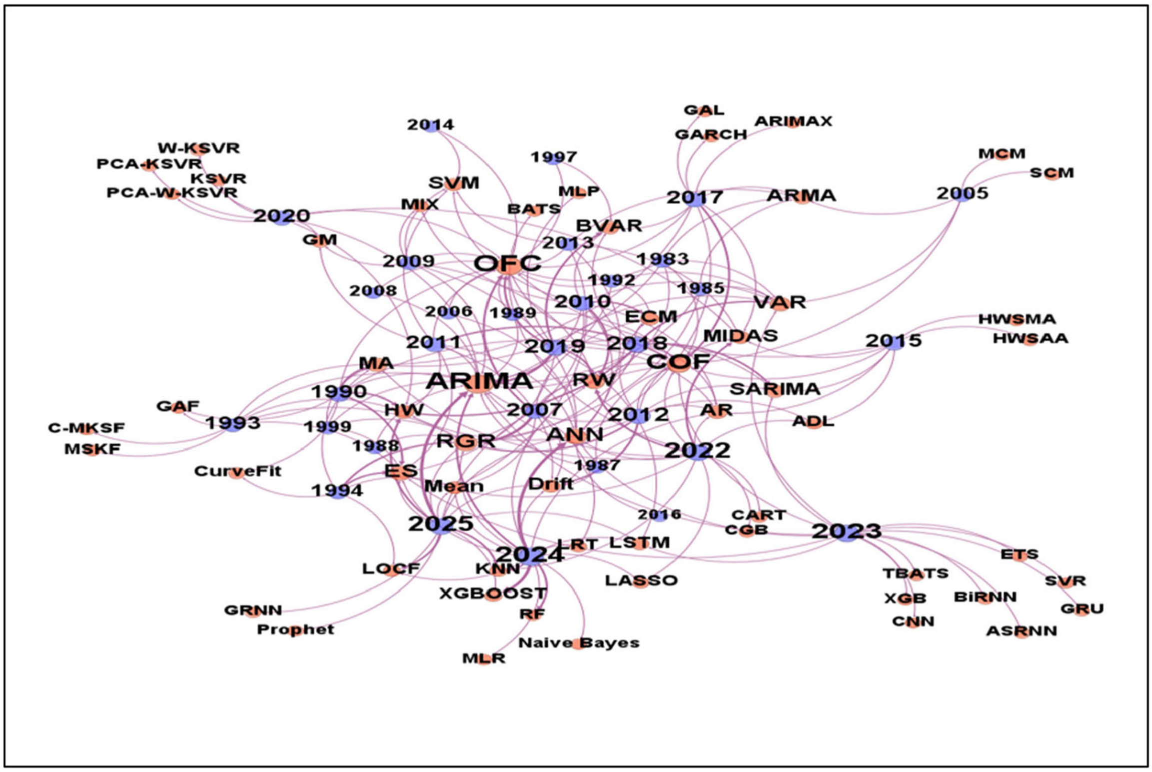


FIGURE 1 | Evolution of forecasting methods in budgeting literature over time.

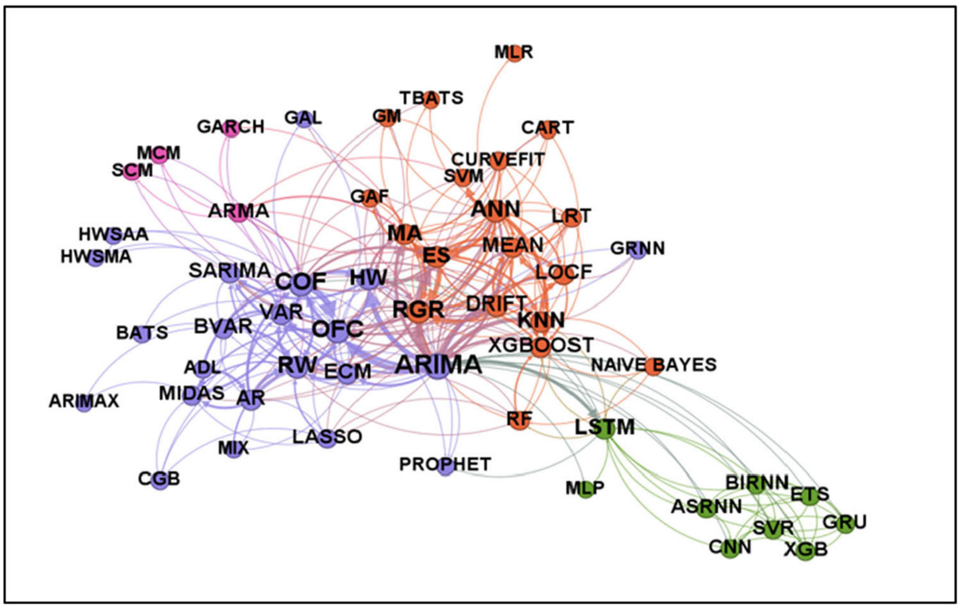
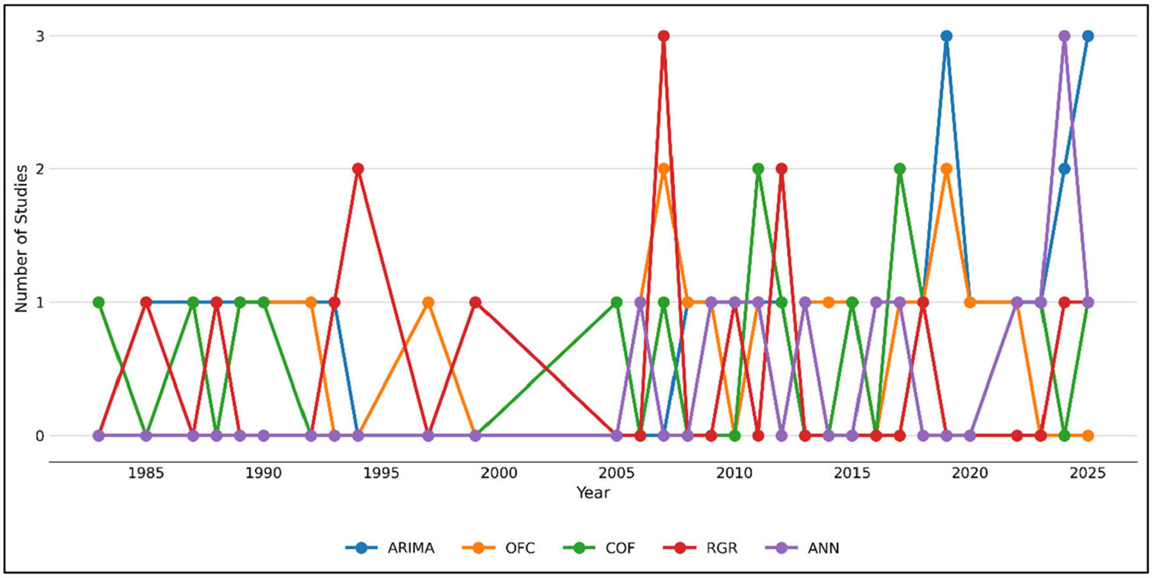


FIGURE 2 | Co-occurrence of forecasting methods in comparative budget studies.

references. The ARIMA model, however, has remained a consistently popular tool for comparison since the 1990s. Regression-based methods experienced a resurgence in the early 2000s, but their popularity subsequently declined somewhat. ML methods, such as ANN, have increasingly featured in comparative studies since the 2010s. COF, on the other hand, has maintained a stable presence among the methods consistently used for comparison throughout the entire period under review. These trends indicate that the literature has preserved its reference points based on traditional methodologies while gradually transitioning toward new-generation, AI-based approaches. Overall, the current methodological spectrum has expanded to encompass both basic statistical models and advanced computational techniques (Tables 2–4).

Fourth, the structural characteristics of the datasets used in the training of forecasting models in the reviewed studies have been analyzed. This analysis is critical for understanding the capacity and diversity of the data sources feeding the models. Below, an assessment is presented showing the distributions of the training datasets in terms of time span (number of years),



GRAPH 1 | Trends in the popularity of key forecasting methods over time (best 5).

TABLE 2 | Characteristics of training datasets in budget forecasting studies.

Period	Studies (*N*)	Observation range	Studies (*N*)	Frequency	Studies (*N*)
1–5 years	4	1–25	8	Quarterly	28
6–10 years	13	26–50	16	Annual	26
11–15 years	17	51–75	10	Monthly	23
16–20 years	10	76–100	6	Biannual	1
21–25 years	9	101–150	12		
26–30 years	2	151–250	7		
31–35 years	7	251–400	3		
35–50 years	4	401–1000	4		
50+ years	3	1000+	3		

TABLE 3 | Country-level distribution of budget forecasting research.

Countries	Studies (*N*)
USA	30
China	6
Panel	6
Türkiye	4
Bharat	2
Brazil	2
South Africa	2
Others*	17

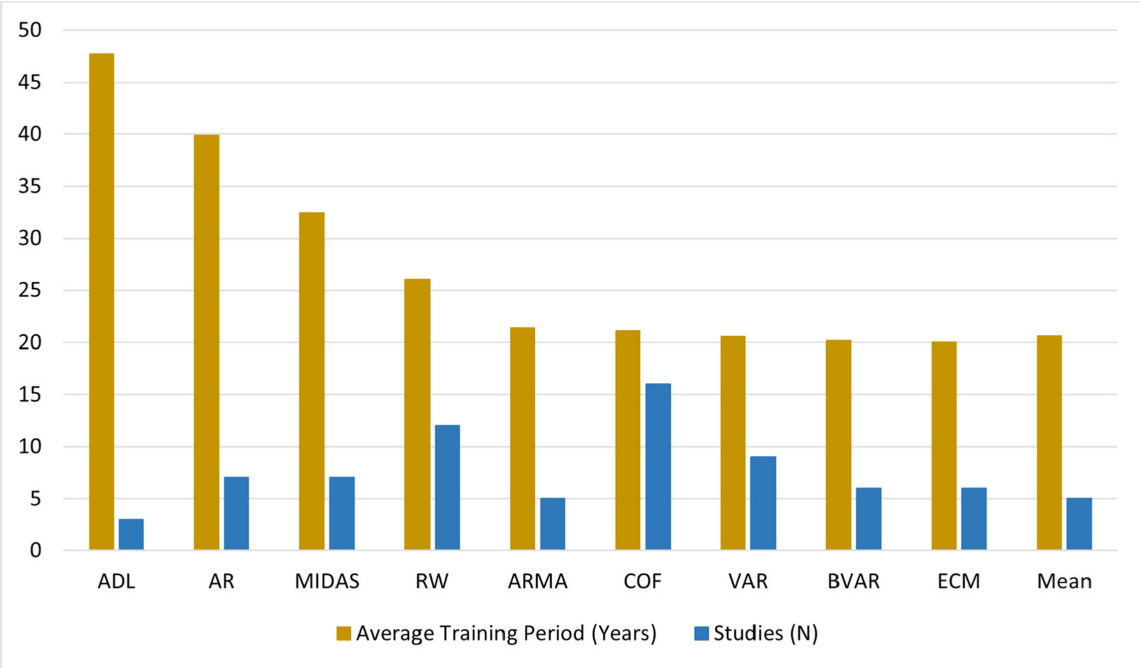
*17 different countries studied once.

TABLE 4 | Number of studies using each evaluation metric.

Metric	Frequency
MAPE	38
RMSE	36
MAE	22
MPE	7
PE	6
MSE	5
ME	5
Others	9
None	2

number of observations, and the most commonly used data frequencies (Graph 2).

Examining the distribution of the time spans of the training datasets, it is evident that the vast majority of studies (over 60%) rely on datasets covering 6–25 years. Notably, datasets spanning 11–15 years are the most commonly used period (17 studies), indicating a clear tendency in the literature toward a medium-term forecasting horizon. This approach allows for capturing macroeconomic cycles by maintaining sufficient historical depth



GRAPH 2 | Average training data span by forecasting model (best 10).

while limiting potential parameter instability issues associated with very long time series. On the other hand, studies using datasets of 31 years or more are relatively few (14 studies in total), reflecting both the accessibility challenges and methodological risks posed by structural breaks in very long series. Conversely, the small number of studies (4 studies) relying solely on short-term data of 1–5 years can be considered characteristic of contexts with limited data depth, particularly in developing economies.

In terms of the number of observations, the literature exhibits considerable diversity. A significant portion of studies is concentrated in the 26–50 observation range (16 studies), while those based on more than 100 observations also constitute a notable group (28 studies). However, only 16 of these 28 studies contain 150 or more observations, indicating that studies relying on long series remain relatively limited in the literature. The presence of eight studies with only 1–25 observations raises serious concerns regarding the reliability of the models used. Forecasting based on such short series entails substantial methodological weaknesses, particularly in terms of testing the statistical significance of parameters and producing robust forecasts. This issue may prevent data-intensive methods, such as DL, from being fully evaluated, highlighting varying levels of reliability in the literature and necessitating cautious interpretation of results from studies with limited observations.

Examining data frequencies, it is observed that quarterly (28 studies), annual (26 studies), and monthly (23 studies) data are chosen with relatively similar frequency. This distribution indicates that, rather than a concentration on a single frequency, different frequencies are used at comparable rates. Accordingly, researchers' choices are largely shaped by data availability, the scope of the study, and the intended forecasting horizon.

The findings on the average training periods of models in Figure 3 show significant alignment when considered alongside the tabular data. Notably, studies based on relatively simple models such as ADL and AR have average training periods exceeding 40 years, indicating that long series are necessary to ensure the parametric stability of these models. In contrast, more complex models such as MIDAS or ARMA can be applied with comparatively shorter datasets, demonstrating the adaptability of methods to series of different lengths. However, it is noteworthy that the relationship between model type and number of observations does not follow a linear pattern. While some short-term studies correspond to low-complexity models, long-term series tend to be associated with models sensitive to parametric stability. This suggests that the alignment between model choice and training data length is shaped not only by technical considerations but also by data accessibility and the objectives of the study.

Fifth, the average forecasting horizons of the methods used in the reviewed studies have been analyzed. This analysis provides important insights into which time spans are preferred for different methodological approaches and whether certain methods are characteristically associated with short- or long-term forecasts.

The findings in Graph 3 indicate a clear clustering of methods in terms of forecasting horizons. Traditional statistical methods, such as AR and MA models, stand out with long average horizons of 14.3 and 12 years, respectively. Similarly, the MIDAS method, with an average horizon of 9.7 years, is among the methods used for long-term forecasts. This suggests that relatively simple linear methods tend to provide a more stable option for long-term forecasts.

In contrast, parametric and multivariate models tend to focus on shorter forecasting horizons. For example, ARIMA (5.5 years), VAR (5.1 years), and BVAR (4.7 years) are predominantly applied in short- to medium-term forecasts. This pattern indicates that as the number of parameters and the modeling

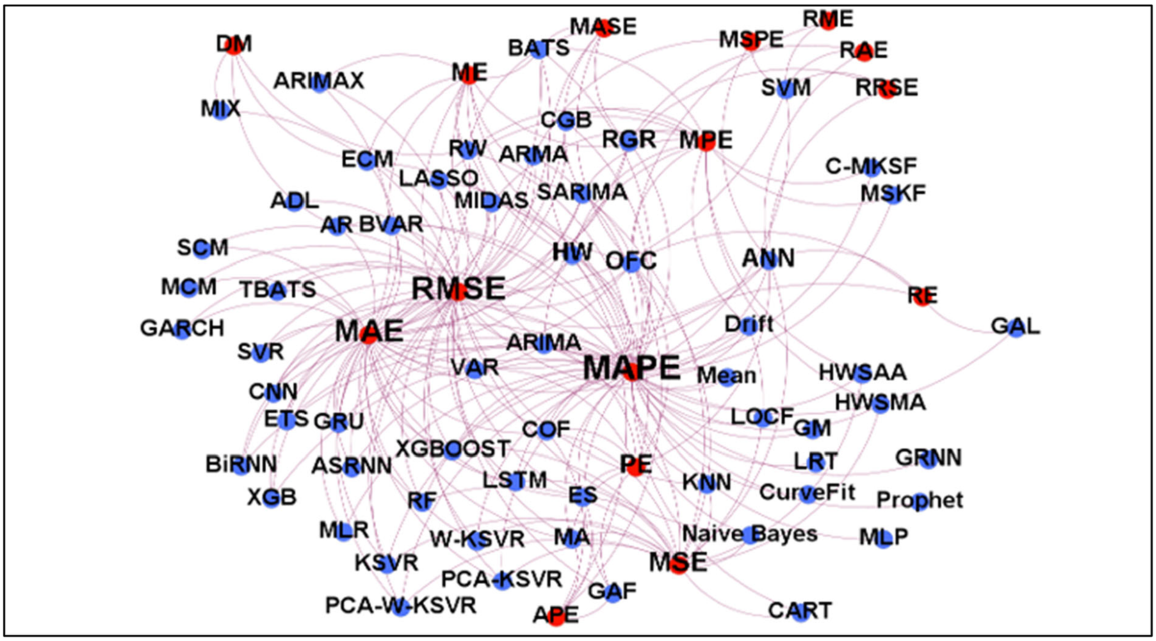
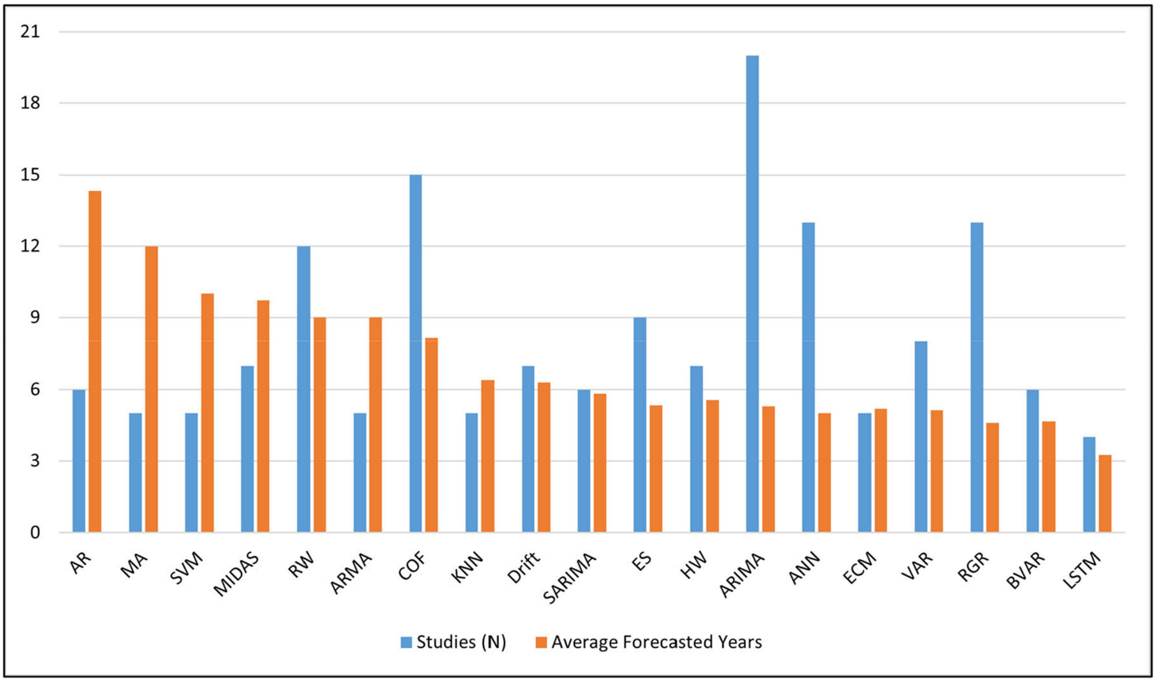


FIGURE 3 | Relationship between methods and performance metrics.



GRAPH 3 | Average forecasting horizons by method in reviewed studies.

flexibility of these models increase, their stability diminishes in the face of long-term uncertainty.

In ML- and DL-based methods, the forecasting horizon is notably shorter. ANN stands out with an average horizon of 5.3 years, while LSTM is limited to only 3.3 years. These results indicate that such methods are strong in capturing complex, nonlinear relationships over short-term periods but are more constrained in long-term forecasts due to risks of parameter instability and overfitting.

Overall, a methodological division of labor emerges among forecasting methods: simple linear statistical methods are concentrated in relatively long-term forecasts, parametric and multivariate models in medium-term horizons, and ML and DL methods in short-term forecasts. This distribution indicates that forecasting horizons are determined not only by researchers' preferences but also by the structural limitations inherent in each method.

Sixth, the geographic distribution of the reviewed comparative studies has been analyzed. This analysis provides important insights into which countries and regions academic research on budget forecasting methods is concentrated, thereby shedding light on the geographic coverage of the existing literature.

The findings on country distribution reveal a pronounced concentration and imbalance in the literature. The United States accounts for the largest share with 30 studies, followed by China (6), panel datasets (6), and Turkey (4). Brazil, India, and South Africa are represented by two studies each. The remaining 17 countries are covered only once and are grouped under the "Others" category. This pattern indicates that empirical research is heavily focused on a limited set of countries, while a wide range of other nations receives very limited attention.

The reasons for the United States' clear predominance include easy access to data, long historical macroeconomic series, the perception of U.S. data as a global benchmark in academic publications, and the dominance of Anglo-Saxon literature. In contrast, for developing countries, both difficulties in accessing long-term reliable data series and differences in institutional contexts result in their underrepresentation in studies. Turkey's representation by only four studies serves as a concrete example of this overall imbalance.

On the other hand, the presence of six studies using panel datasets indicates that researchers have also pursued cross-country comparisons rather than focusing solely on single-country analyses. However, the limited number of panel studies suggests that the literature has not yet fully developed a comparative perspective capable of providing generalizable insights and revealing cross-country differences.

Seventh, the distribution of metrics used to evaluate forecasting performance in the reviewed comparative studies has been analyzed. This analysis provides important insights into which evaluation criteria are prioritized in the literature and, consequently, which dimensions of error are considered most critical in method comparisons.

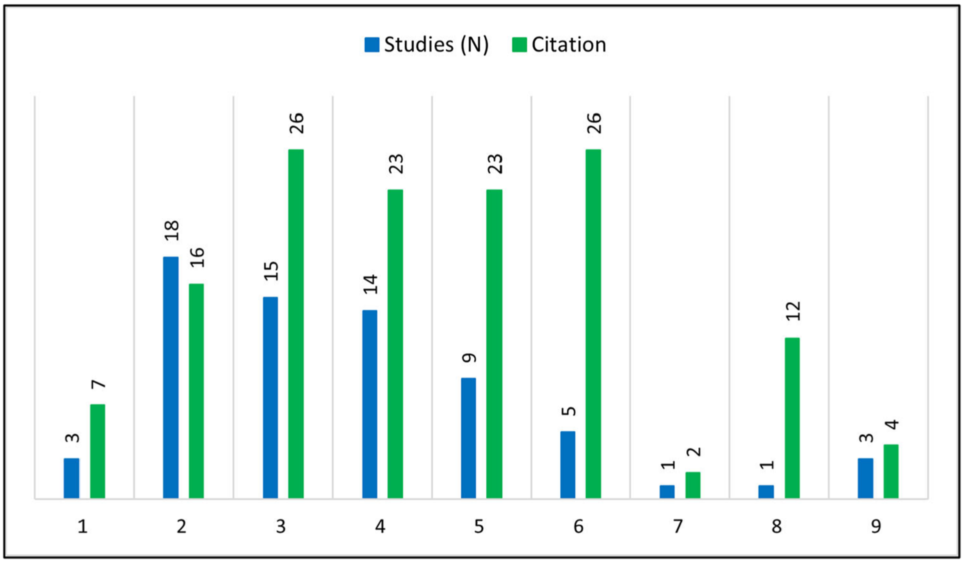
Examining the metrics used to evaluate method performance, a clear concentration is observed in the literature. The most frequently used evaluation criteria are MAPE (38 studies), RMSE (36 studies), and MAE (22 studies). These three metrics account for the vast majority of usage, establishing a dominant position in assessing forecasting accuracy. The prominence of MAPE and RMSE can be explained by their ease of interpretation and their suitability for comparisons across series of different magnitudes.

In contrast, metrics such as MPE (7 studies), ME (5 studies), MSE (5 studies), and PE (6 studies) are used relatively infrequently. Their limited representation indicates that the literature tends to favor standardized metrics that allow for international comparisons. Additionally, in nine studies categorized as "Others," more specific or context-specific metrics were employed, while two studies did not report any performance metric at all. The latter case can be regarded as a significant shortcoming in terms of methodological transparency.

The network structure in the graph further clarifies the relationships between methods and metrics. RMSE and MAPE stand out as nodes with the most extensive connections, showing strong associations with widely used methods such as ARIMA, ANN, HW, VAR, OFC, COF, and SARIMA. This finding indicates that performance evaluations of prominent methods in the literature generally rely on the same set of metrics, and comparability is largely established around these measures. In contrast, less frequently used methods such as LSTM, XGBoost, or Prophet are linked to a relatively limited number of metrics, suggesting both low methodological diversity and the absence of a standardized approach for evaluating the performance of these methods.

Eighth and finally, the relationship between method diversity and academic impact in the reviewed comparative studies has been analyzed. This analysis provides important insights into whether the number of methods compared in a study has any effect on the number of citations the study receives (Graph 4).

The analysis results indicate a notable but nonlinear relationship between the number of methods in a study and its citation count. The vast majority of the reviewed studies (over 70%) compare between 2 and 5 methods. Studies with 2 methods



GRAPH 4 | Relationship between studies, methods, and citations.

represent the most common category with 18 cases, followed by 3-method studies with 15 cases and 4-method studies with 14 cases. In terms of citation counts, the highest average citation performance is observed among studies comparing between 3 and 6 methods. Studies with 3 methods receive an average of 26 citations, while those with 6 methods reach a similar average of 26 citations. This finding suggests that a moderate level of method diversity may enhance academic impact.

However, it is also evident that citation performance does not increase consistently with the number of methods. Studies including 7 or more methods exhibit a noticeable decline in citation counts. This suggests that comparisons involving an excessive number of methods may be less favored by readers and researchers or that such studies may give the impression of emphasizing methodological breadth over analytical depth. Interestingly, studies employing only a single method are quite rare (3 studies), and their average citation count (7 citations) remains relatively low. This finding suggests that non-comparative studies may have a more limited academic impact compared to comparative studies.

In conclusion, this analysis suggests the existence of an optimal range for method diversity. Comparisons with very few methods are considered insufficient, while studies including an excessive number of methods may be less favored by readers. Studies comparing between 3 and 6 methods exhibit the highest academic impact, indicating that comprehensive yet focused comparisons contribute more valuable insights to the literature.

5 | Discussion and Policy Recommendations

The comprehensive analysis presented in this study offers a multidimensional examination of the comparative structure of the budget forecasting literature, characterized by methodological diversity, geographic biases, and the ongoing tension between model complexity and practical applicability within a dynamic and evolving field. The following discussion synthesizes these findings to draw broader implications, address inherent tensions in the field, and suggest directions for future research and practice.

5.1 | Methodological Diversity and the Quest for the Optimal Model

The most striking finding of this study is the relentless diversification of forecasting methods. The evolution from simple ARIMA models to sophisticated LSTM networks and hybrid PCA-W-KSVR frameworks represents a true methodological arms race. This expansion is undoubtedly fueled by the quest for accuracy, supported by increasing computational power and richer data availability. However, the analyses indicate that this proliferation does not yield a universally superior algorithm; rather, it has led to a form of methodological inflation that can overwhelm practitioners and obscure clear guidance. The absence of a single “best” method should not be viewed as a failure of the research community but as a reflection of the inherent complexity of budgetary processes. Budget data are the product not of a single, static process, but of political negotiations, economic shocks, institutional changes, and administrative rules. Accordingly, the focus should shift from finding the optimal model to identifying the right model for the right context.

5.2 | The Enduring Role of Simple Models and the Benchmarking Paradox

Despite the influx of complex techniques, simple models such as OFC, RW, and various regression approaches remain not only relevant but also serve as fundamental benchmarks. Any new model must consistently outperform these naïve or simple standards to justify its added complexity. The fact that these simple models still frequently rank among the top performers, or at least are not consistently the worst, in comparative studies serves as a cautionary reminder for the field. It underscores that statistical sophistication does not automatically translate into forecasting superiority, particularly in contexts where data are noisy, short, and subject to unpredictable political interventions. This creates a paradox: as research energy increasingly focuses on developing complex models, the practical value of these advancements often remains marginal when compared against well-defined, simple benchmarks.

5.3 | Data Quality, Quantity, and the Global Inequality Issue

The analysis of training data characteristics highlights a significant and often overlooked challenge: data constraints. The prevalence of studies using 11–15 years of data indicates that many models are built on relatively limited information sets. This poses a critical limitation for data-hungry methods such as DL, potentially leading to overfitting and reduced out-of-sample generalization performance. The geographic distribution of studies further exposes a substantial data gap. The dominance of United States-based research reflects not merely academic preference but is directly linked to data availability, quality, and length. The underrepresentation of developing economies constitutes a critical gap in the literature. Budgetary processes, institutional frameworks, and economic fluctuations in these contexts differ substantially. Models optimized on long, stable U.S. time series may fail when applied to economies prone to fiscal shocks, characterized by lower data quality and shorter historical records.

5.4 | Evaluation Inconsistency and the Hegemony of Metrics

The near-absolute dominance of MAPE, RMSE, and MAE as performance metrics provides a common language for comparison but also poses a potential pitfall: the tyranny of averages. These metrics measure the magnitude of errors but remain largely agnostic to the direction and timing of mistakes. In a budgetary context, the cost of overestimating a deficit (potentially leading to unnecessary austerity measures) is substantially different from the cost of underestimating it (potentially triggering a fiscal crisis). The near-complete neglect of directional metrics such as MPE in the current literature represents a significant shortcoming. Evaluations should incorporate asymmetric loss functions that reflect the real-world political and economic costs of errors, rather than solely their statistical properties.

5.5 | Academia–Practice Gap

Citation impact analysis provides a striking insight into the sociology of the field. The higher citation rates of studies comparing 3–6 methods suggest that the academic community values comprehensive yet focused comparisons. However, this academic ideal does not always align with practical needs. A practitioner in a treasury department does not require a comparison of six different ML models; they need a single, robust, interpretable, and operationally feasible model tailored to their specific national context. This points to a potential academia–practice disconnect. The complexity of state-of-the-art models (e.g., LSTMs and XGBoost) often renders them opaque "black boxes" for policymakers. Simpler models or official forecasts, even if slightly less accurate by some metrics, offer transparency and accountability. Future research should address this gap by not only reporting accuracy metrics but also considering operational aspects, such as computational requirements, interpretability, and integration into existing budgetary processes.

5.6 | Rise of Hybrid and Ensemble Approaches

The emergence of hybrid models points, albeit modestly, toward a promising direction for the field. Rather than treating methods as competitors, the discipline is gradually recognizing the value of hybridization and ensemble techniques. The strengths of one model can compensate for the weaknesses of another. For instance, a classical statistical model may capture long-term trends, while an ANN can adjust for recent nonlinear deviations. Combining forecasts from different model families often yields more accurate and robust results than any single model. This meta-modeling approach, which focuses on integrating information rather than selecting a single "winner," appears to be a more efficient pathway than developing isolated, increasingly complex algorithms.

6 | Conclusion

This study systematically analyzes the literature examining the comparative performance of budget forecasting methods, providing a comprehensive overview of the field's methodological evolution, geographic distribution, data characteristics, and performance evaluation trends. The findings indicate that budget forecasting is not a static domain but a dynamic research area in continuous evolution, deeply influenced by developments in related disciplines.

The analyses of this study clearly demonstrate that there is no single, universally optimal method for budget forecasting. Instead, the performance of forecasting methods varies substantially depending on contextual factors such as economic conditions, data quality, institutional structures, and forecast horizon. The resilience of traditional statistical methods in long-term forecasts, alongside the strength of ML and DL approaches in capturing short-term complex relationships, indicates the existence of a methodological division of labor.

The pronounced geographic concentration in the literature, particularly the underrepresentation of emerging economies, poses significant limitations to the generalizability of existing findings. Similarly, the near-absolute dominance of metrics such as MAPE, RMSE, and MAE in performance evaluation leads to the neglect of critical dimensions, including the direction of errors and asymmetric costs.

The findings of this study offer important implications for future research. First, contextual appropriateness should take precedence over methodological complexity. Second, the geographic scope should be expanded to include emerging economies, and datasets should be enriched accordingly. Third, performance evaluation criteria should be diversified to reflect the policy costs of forecast errors. Finally, hybrid and ensemble learning approaches warrant further investigation to leverage the relative strengths of different methods.

For policymakers and practitioners, this study emphasizes the importance of avoiding dogmatic method selection in budget forecasting. Adopting a flexible and pluralistic approach, taking into account institutional capacity, data quality, and forecasting objectives, can contribute to producing more realistic and reliable budget forecasts.

In conclusion, progress in the field of budget forecasting depends less on developing increasingly complex models and more on managing the existing methodological diversity in a systematic and context-sensitive manner. This study provides a comprehensive foundation for such efforts and aims to contribute to the development of budget forecasting as both a science and an art.

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